

10.00 - 01.00 pm.

Instructions to candidates: Question no. 1 is compulsory; solve any five from Q.2 to Q.7. If any data is missing is a part of question. Assume suitable data giving reason.

Q1. Fill in the blanks with suitable word/s.

[1×10=10marks]

- The newton raphson method requires memory and time as compare to guassseidal method for solution of load flow.
- The purpose of fault analysis is to determine and selection of appropriate
- The determination of original phasors in terms of sequence component is termed as
- Phase shifting transformer are used to control and is used for estimating
- The major functions of an excitation system are 1 2 and 3
- The transient stability of the generator is dependent on the fault and type.
- A turbine with multiple sections may be either or
- To provide damping, the stabilizer must produce a component of electrical torque in phase with the
- In PSS the signal washout block serves as a
- The input signal to the excitation system is normally the generator

Q2.

[2×6=12marks]

- Explain modeling of phase shifting and regulating transformer
- Write a short note on modeling of lighting arrester

Q3.

[2×6=12marks]

- Enlist desirable features of an ideal load flow method. Compare merits and demerits of guassseidal method.
- Derive swing equation for transient stability.

Q4.

[2×6=12marks]

- How equal area criterion is helpful to calculate transient stability?
- Two 11kV, 20 MVA, three phase, star connected generators operate in parallel. The positive, negative and zero sequence reactance of each being, respectively, $j0.18$, $j0.15$, $j0.10$ pu. The star point of one of the generators is isolated and that of the other is earthed through a 2.0 ohm resistor as shown in fig.4.b. Assume that the grounded generator is solidly grounded. Find the fault current in each phase and voltage of the healthy phase for a double line to ground fault on terminals of the generator. Assume solid fault ($Z^f = 0$).

Q5.

[2×6=12marks]

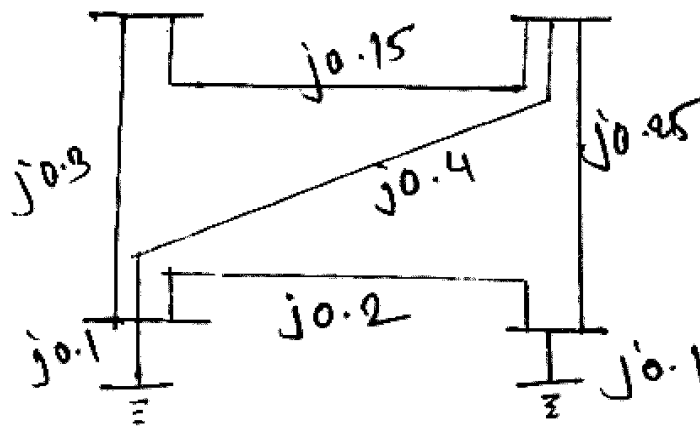
- From fundamentals explain steam turbine modeling.

The following data relate to reheater steam volume

$$Q_o = 230 \text{ kg/s}, V = 115 \text{ m}^3, P_o = 4140 \text{ kpa}, \frac{\partial \rho}{\partial P} = 0.0035$$

Calculate the reheater time constant.

- Formulate Z_{bus} matrix for given power system.



Q6.

- What are the effects of various types of disturbance parameters on stability?
- With block diagram discuss on requirement of excitation control system.

Q7.

- What is the need of power system stabilizer in system stability?
Draw block diagram of power system stabilizer and explain.
- Discuss various methods for augmentation of stability.

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